



Customer Shopping Behavior Analysis

Analyzing 3,900 transactions to uncover spending patterns, customer segments, and product preferences — driving strategic business decisions.

Dataset at a Glance

3,900

Transactions

Total purchase records

18

Features

Columns per record

37

Missing Values

Review Rating column

50

Locations

Unique geographic areas

Key features include customer demographics, purchase details (category, amount, season), and shopping behavior (discounts, subscription status, review ratings).



Data Preparation in Python

→ Clean & Standardize

Checked nulls, imputed missing Review Ratings using median by category, renamed columns to snake_case.

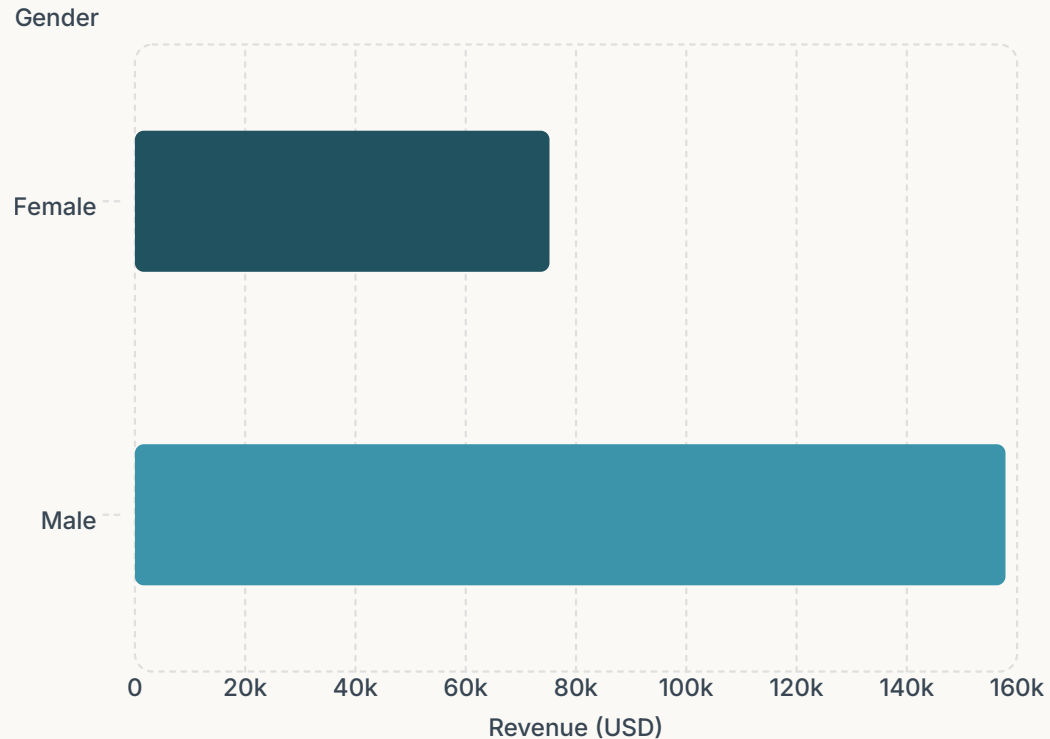
→ Feature Engineering

Created **age_group** bins and **purchase_frequency_days**; dropped redundant promo_code_used column.

→ Database Integration

Loaded cleaned DataFrame into PostgreSQL for SQL-based analysis.

Revenue by Gender



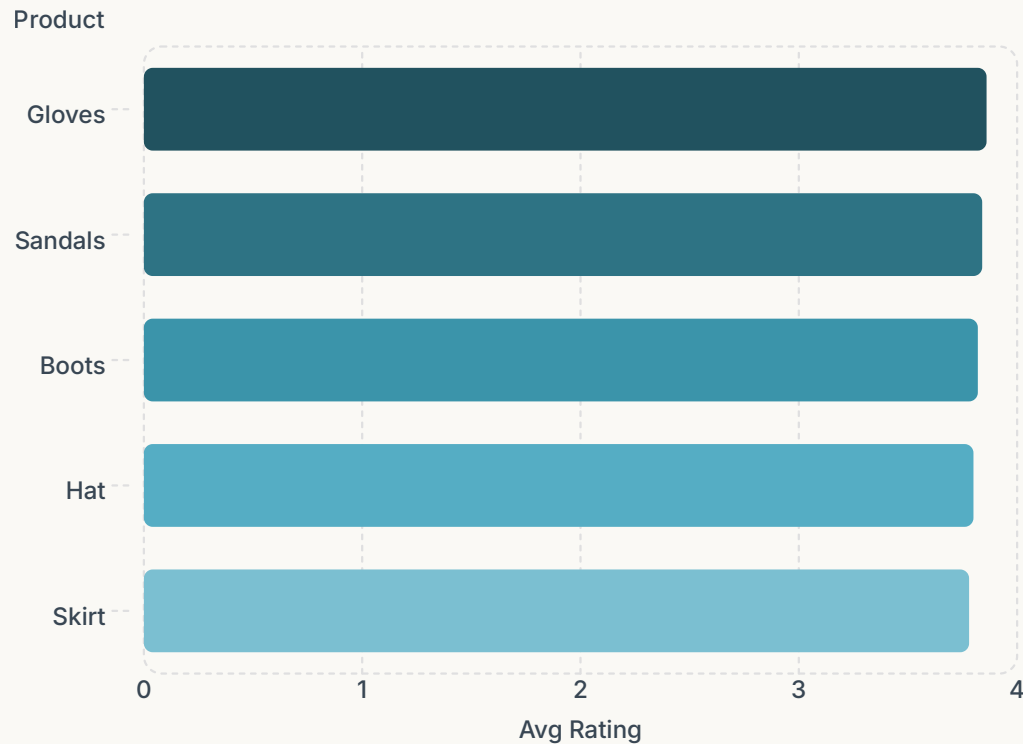
Key Insight

Male customers generated **\$157,890** in revenue — more than double the **\$75,191** from female customers, despite females representing a significant portion of buyers.

- ❗ This disparity suggests an opportunity to increase engagement and average spend among female customers.

Product & Discount Insights

Top 5 Products by Rating

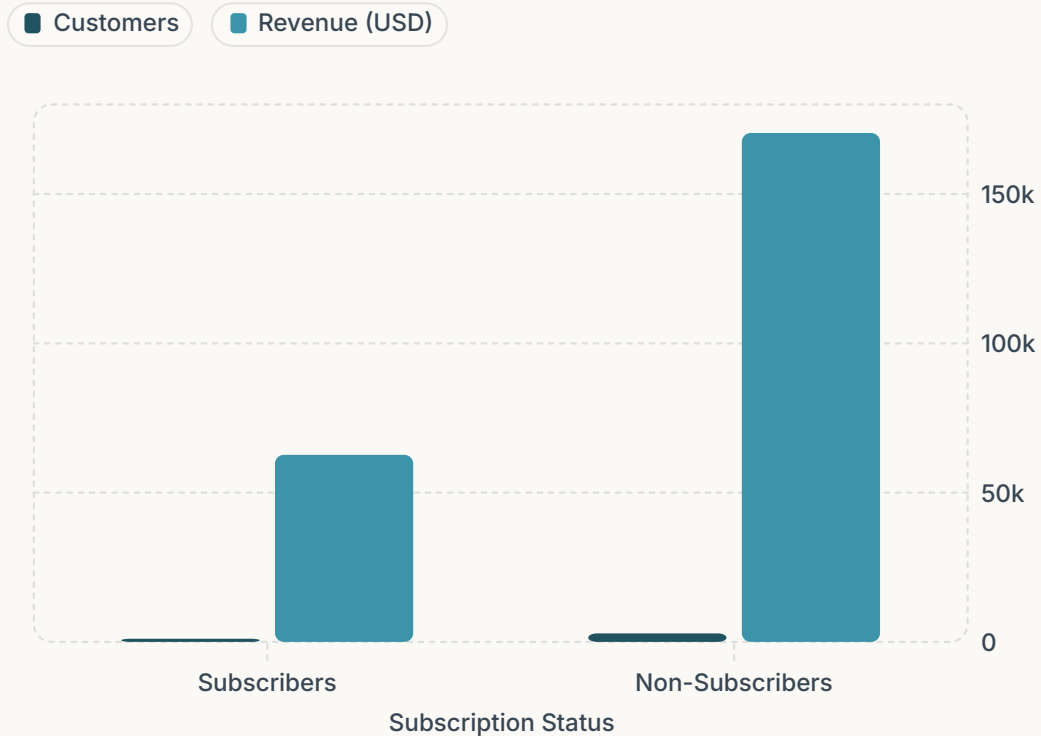


Most Discount-Dependent Products

- **Hat** — 50% of purchases used a discount
- **Sneakers** — 49.66%
- **Coat** — 49.07%
- **Sweater** — 48.17%
- **Pants** — 47.37%

839 high-spending customers used discounts and still spent above average purchase amount.

Subscription & Shipping Analysis



Key Findings

- **73%** of customers are non-subscribers, generating **\$170,436** in revenue
- Subscribers (27%) average **\$59.49/purchase** vs. **\$59.87** for non-subscribers
- Express shipping users spend slightly more on average (**\$60.48**) vs. Standard (**\$58.46**)
- Of repeat buyers (>5 purchases), **958** subscribe vs. **2,518** who don't

Customer Segmentation & Age Group Revenue

Loyal

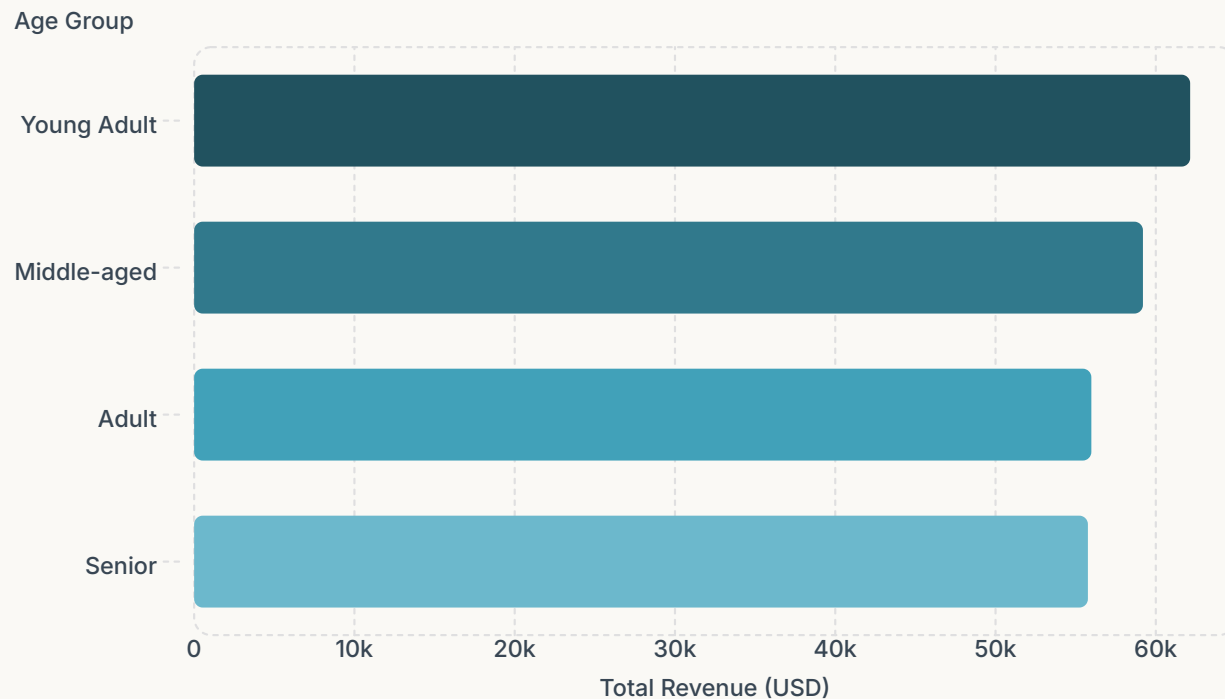
3,116 customers — the dominant segment

Returning

701 customers — growth opportunity

New

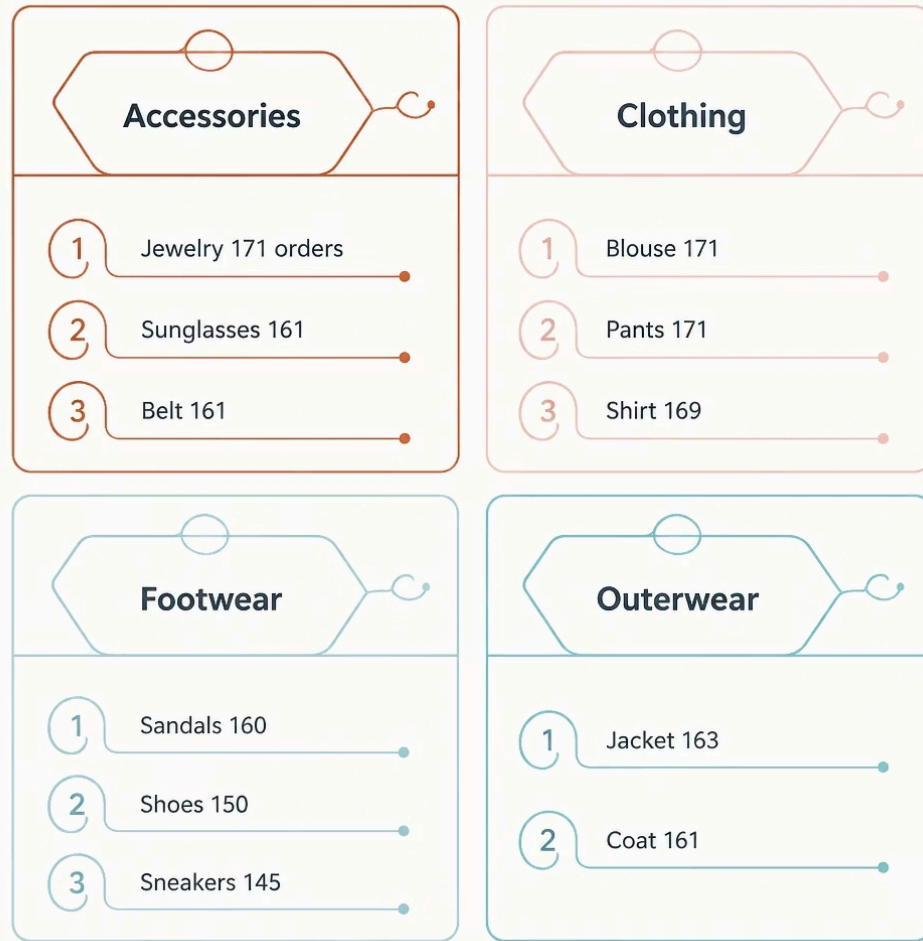
83 customers — smallest segment



Young Adults lead revenue contribution at **\$62,143**, with a gradual decline across older age groups.



Top Products by Category



Category Highlights

- **Clothing** dominates with Blouse and Pants each at 171 orders
- **Outerwear** led by Jacket (163) and Coat (161)
- **Footwear** — Sandals top the category with 160 orders
- **Accessories** — Jewelry leads with 171 orders

Clothing is the highest-revenue category overall, followed by Accessories, Footwear, and Outerwear.

Power BI Dashboard

3.9K

Customers

\$59.76

Avg Purchase

3.75

Avg Rating

73%

Non-Subscribers

The interactive dashboard visualizes subscription status, revenue and sales by category, and performance across age groups – enabling quick, data-driven decisions.

Business Recommendations



Boost Subscriptions

Promote exclusive subscriber benefits to convert the 73% non-subscriber base.



Loyalty Programs

Reward repeat buyers to advance them into the Loyal segment.



Review Discount Policy

Balance sales boosts with margin control — 50% of Hat purchases use discounts.



Targeted Marketing

Focus on Young Adults and express-shipping users for highest revenue impact.